

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights such as spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.

- **Feature Engineering:**

- Created age_group column by binning customer ages.
- Created purchase_frequency_days column from purchase data.

- **Data Consistency Check:** Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in **MySQL** to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.
2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.
3. **Top 5 Products by Rating** – Found products with the highest average review ratings.
4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.
5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.
6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.
7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.
8. **Top 3 Products per Category** – Listed the most purchased products within each category.
9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.
10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

Findings and recommendations

1. In the **clothing category**, adult and young adult Customers prefer debit & credit card options over other payment modes 18-20% more than middle-aged and senior customers. **Recommendation - credit and debit card offers based marketing** to them would help in maximizing sales.
2. **Customer Loyalty Programs** – Reward repeat buyers and customers with subscription to move them into the “Loyal” segment.
3. All the customers **with subscriptions** fully depend on discount- **review discount policy & margins**.
4. Average purchase frequency of clothing & accessories is **90 days/ Target 80-85 days** Recommendation- **Promote new arrivals** every 6-8 weeks.

